

Администрирование локальных сетей

Лабораторная работа №6

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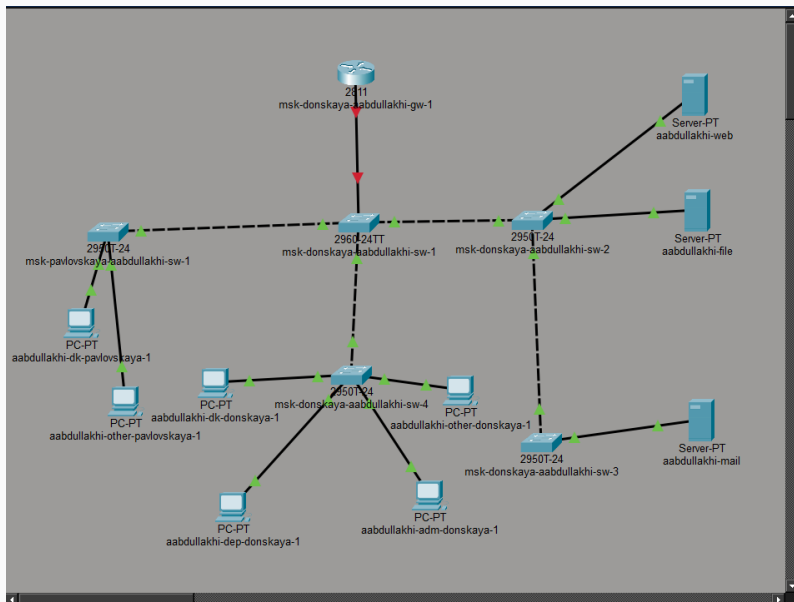
27 февраля 2026

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Цель

Целью данной работы является изучение и практическая настройка статической межсетевой маршрутизации между виртуальными локальными сетями (VLAN) в среде эмуляции сети Cisco Packet Tracer.

Выполнение лабораторной работы



Первичная настройка маршрутизатора

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-aabdullakhi-gw-1
msk-donskaya-aabdullakhi-gw-1(config)#line vty 0 4
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#console 0
^
% Invalid input detected at '^' marker.

msk-donskaya-aabdullakhi-gw-1(config-line)#line console 0
msk-donskaya-aabdullakhi-gw-1(config-line)#password cisco
msk-donskaya-aabdullakhi-gw-1(config-line)#login
msk-donskaya-aabdullakhi-gw-1(config-line)#enable secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#service password-encryption
msk-donskaya-aabdullakhi-gw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-aabdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-aabdullakhi-gw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-aabdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-aabdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:5:20.970: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:5:20.970: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-aabdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-aabdullakhi-gw-1(config-line)#
```

Рис. 2: настройка маршрутизатора

Настройка маршрутизации между VLAN

Physical Config CLI Attributes

```
msk-donskaya-aabdullakhi-gw-1(config-line)#
msk-donskaya-aabdullakhi-gw-1(config-line)#exit
msk-donskaya-aabdullakhi-gw-1(config)#interface f0/0
msk-donskaya-aabdullakhi-gw-1(config-if)#no shutdown

msk-donskaya-aabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-if)#interface f0/0.2
msk-donskaya-aabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-subif)#encapsulation dot1q 2
msk-donskaya-aabdullakhi-gw-1(config-subif)#ip address 10.128.1.1 255.255.255.0
msk-donskaya-aabdullakhi-gw-1(config-subif)#description managment
msk-donskaya-aabdullakhi-gw-1(config-subif)#description management
msk-donskaya-aabdullakhi-gw-1(config-subif)#
msk-donskaya-aabdullakhi-gw-1(config-subif)#interface f0/0.3
msk-donskaya-aabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-subif)#encapsulation dot1q 3
msk-donskaya-aabdullakhi-gw-1(config-subif)#ip address 10.128.0.1 255.255.255.0
msk-donskaya-aabdullakhi-gw-1(config-subif)#description management
msk-donskaya-aabdullakhi-gw-1(config-subif)#description servers
msk-donskaya-aabdullakhi-gw-1(config-subif)#description servers
msk-donskaya-aabdullakhi-gw-1(config-subif)#interface f0/0.101
msk-donskaya-aabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.101, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

msk-donskaya-aabdullakhi-gw-1(config-subif)#encapsulation dot1q 101
msk-donskaya-aabdullakhi-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-aabdullakhi-gw-1(config-subif)#description dk
msk-donskaya-aabdullakhi-gw-1(config-subif)#
msk-donskaya-aabdullakhi-gw-1(config-subif)#interface f0/0.102
msk-donskaya-aabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down
```

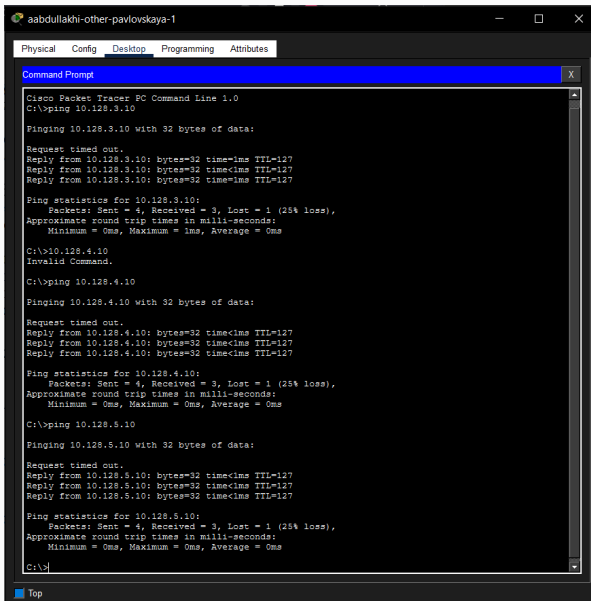
Активация Windows

Чтобы активировать Windows, перейдите в "Параметры".

Copy

Paste

Проверка связности



```
aabdullakhi-other-pavlovskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.10

Pinging 10.128.3.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.4.10
Invalid Command.

C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.10

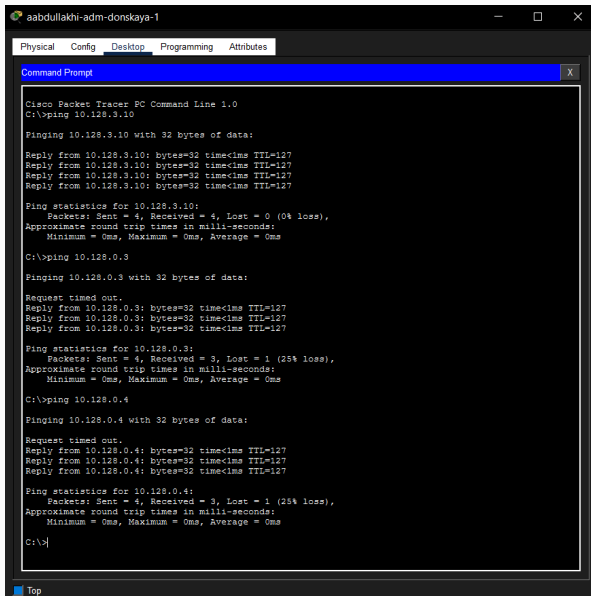
Pinging 10.128.5.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```


Дополнительная проверка



```
aabdullakhi-adm-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.10

Pinging 10.128.3.10 with 32 bytes of data:

Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.3

Pinging 10.128.0.3 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.3: bytes=32 time<1ms TTL=127
Reply from 10.128.0.3: bytes=32 time<1ms TTL=127
Reply from 10.128.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Top

Анализ трафика в режиме Simulation

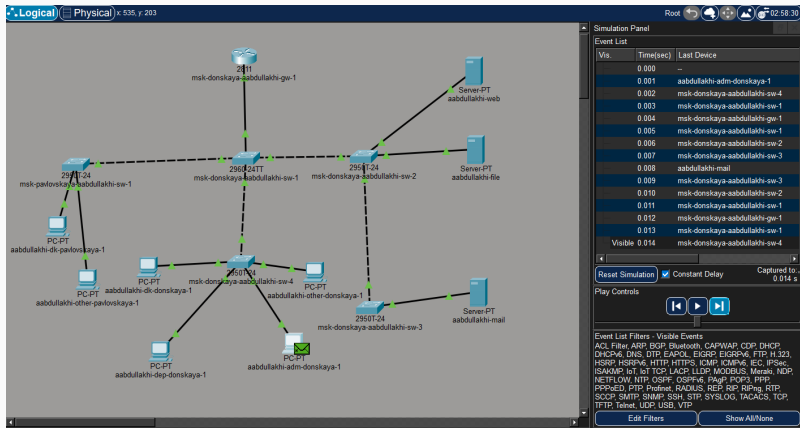
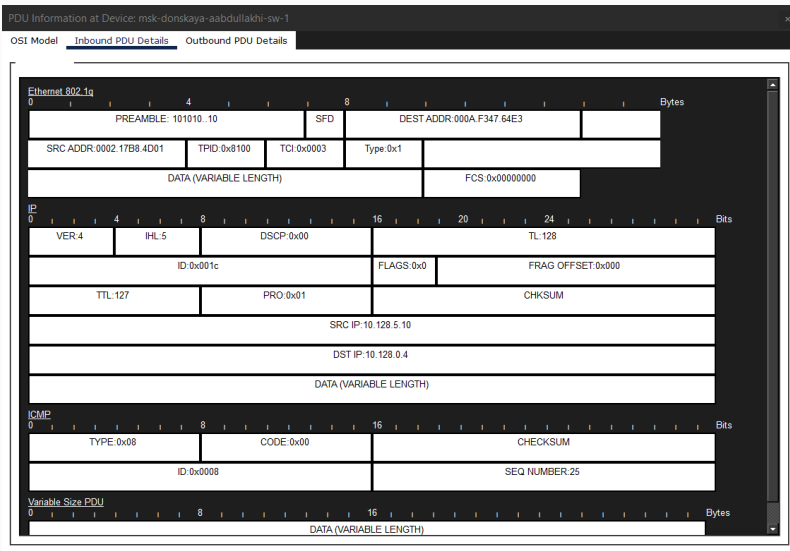


Рис. 6: Просмотр перемещения пакета в режиме Simulation

Изучение структуры пакета



Выводы

Была выполнена первоначальная настройка маршрутизатора, настроен удалённый доступ по SSH и создана межсетевая маршрутизация между VLAN с использованием подинтерфейсов. Проверка соединения с помощью команды `ping` показала корректную работу маршрутизации между VLAN. Также был изучен процесс передачи ICMP-пакета и структура сетевых заголовков Ethernet, IP и ICMP.